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WATER QUALITY ASSESSMENT BASED ON GILL MONOGENEAN PARASITES OF *Oreochromis niloticus* (Linnaeus, 1758) AND *Labeo calbasu* (Hamilton, 1822) FISHES OF RIVER PENNA (INDIA)

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ABSTRACT

A year-long (March, 2017 to March, 2018) study targeting the water quality and monogenean parasitosis on the gills of two fish species [*Oreochromis niloticus* (n = 133) and *Labeo calbasu* (n = 122)] was carried out at three different locations of river Penna flowing through YSR Kadapa district, Andhra Pradesh (India). A total of four species of ectoparasitic monogeneans i.e., *Cichlidogyrus sclerosus*, *Cichlidogyrus tilapiae* and *Scutogyrus longicornis* from *O. niloticus* and *Dactylogyrus fotedari* from *L. calbasu* were detected. A significant correlation ($p < 0.05$) was found between the prevalence of monogeneans and some water quality parameters [e.g., temperature, alkalinity, total dissolved solids, total hardness, dissolved oxygen and calcium hardness] in all the fishes examined. Results revealed highest prevalence of monogenean infection during winter, rainy and summer seasons. Parasitosis was analyzed location-wise and fishes collected from Somasila backwaters were observed highly infected with monogeneans. The positive correlation existing between the monogenean infection and water quality variables in the two basic seasons has led to the conclusion that ectoparasitic monogeneans are good biological indicators in assessing the water quality of a water body.

Keywords: Bioindicators, *Labeo calbasu*, monogeneans, *Oreochromis niloticus*, water quality

INTRODUCTION

Monogeneans, the most notorious flatworms (Platyhelminthes), are ectoparasites on various fish species (Hayward, 2005; Tubbs *et al.*, 2005). Their diverse life history traits such as oviparity and viviparity reproductive mechanisms, camouflage and behavioural responses to host and environmental factors have enabled them to become the most successful parasites (Khang *et al*., 2016). In addition, their direct life-cycle has enabled them to be good bioindicators as compared to those having indirect multiple life-cycles such as digeneans (MacKenzie *et al.*, 1995). Several researchers all over the world have acknowledged the monogenean parasites as potential bioindicators of environmental pollution (Marcogliese, 2005; Sanchez-Ramirez *et al*., 2007; Bayoumy *et al.*, 2008; Sures *et al*., 2017) due to their conventional numerical response to most of the water quality variables (Khan and Thulin, 1991). Specifically, their incidence or abundance can illustrate the condition of environment including water quality (Sanchez-Ramirez, 2007; Palm and Ruckert, 2009). Marcogliese *et al*. (1998) observed that monogeneans increase in number at low and medium pollutant concentrations and decrease or disappear at high concentration. The level of pollutants in aquatic systems amplify with constant increase in anthropogenic activities such as industrialization and agricultural revolution.

Hence, it is indispensable to monitor the presence and impact of these pollutants in aquatic systems. Bioindicators may serve as an accurate accumulation- or effect-indicators that can reflect environmental impact due to their ability to respond to habitat variation with changes in physiology or chemical composition of the host (Vidal-Martinez *et al*., 2009). Commendable work has been done on monogenean parasites of fish in relation to water variables in aquatic environment (Vidal-Martinez *et al*., 2009; Modu *et al*., 2014; Adou *et al*., 2017; Lacerda *et al*., 2018). *Oreochromis niloticus* and *Labeo calbasu* are the most widely cultured freshwater fishes and their monogenean Cichlidogyrids and Dactylogyrids are considered to be highly host specific and act as highly sensitive and potential indicators (Sanchez-Ramirez *et al*., 2007). Each of these monogenean species occupy an ecological niche and interact erratically with its environment. The present study was a pilot study conducted on river Penna (Penneru) and was designed to provide some valuable information in understanding the relationship between the occurrence of monogenean parasites and water quality.

MATERIALS AND METHODS

Study area

Penna river is 597 km long which originates from Karnataka state (India), traverses about 61 km before entering into the state of Andhra Pradesh (536 km) and then merges into the Bay of Bengal. Its basin extends over an area of 55,213 km^2 in Peninsular India covering the states of Karnataka and Andhra Pradesh (Fig. 1). Penna river also flows through YSR Kadapa district in Andhra Pradesh touching the three sampling sites.

Fig. 1: Map showing the study area and the flow of river Penna in Kadapa district of Andhra Pradesh (India)

Site-1: Mylavaram reservoir across the Penna river in Mylavaram village (latitude 14°0′15″N, 78°20′40″ E longitude), located in YSR Kadapa district (Fig. 2).

Site-2: Aadinimmayapalle dam across the Penna river in Chennur village (latitude14″34′12″ N, 78″48′0″ E longitude), YSR Kadapa district (Fig. 2).

Site-3: Backwaters of Somasila reservoir across the Penna river in Somasila village (14°29′22″ N 79°18′19″ E) located in SPSR Nellore district, Andhra Pradesh reach near Vontimitta village, YSR Kadapa district (Fig. 2). The main part of river after entering Nellore merges into the Bay of Bengal.

The fishes were exclusively procured from local fishermen at the site of catchment area of river Penna flowing through YSR Kadapa district.

Fish sampling and parasitological assessment

A total of 133 *O. niloticus* fish measuring 5-12 cm (mean: 8.82 ± 1.42 cm) in length and 100-500 g (mean: 229.6 ± 80.7 g) in weight; and 122 *L. calbasu* fish measuring 5-11 cm (mean: 7.95 ± 1.78 cm) in length and 120-1000 g (mean: 429.9 ± 191.42 g) in weight were collected from the above three sampling locations during the study period (March, 2017 to March, 2018). Fish identification was done according to Jayaram (1981) and Munro (1982) and the fishes of various sizes i.e. small, medium and large were brought to the laboratory for parasite examination. Adequate attention was paid to

Site 1: Mylavaram reservoir across the Penna river in Mylavaram village, YSR district

Site 2: Aadinimmayapalle dam across the Penna river in Chennur village, YSR district

Site 3: Backwaters of Somasila reservoir across the Penna river in Somasila village, Nellore district, Andhra Pradesh reach near Vontimitta village, YSR district

Fig. 2: The sites of Penna river selected for fish and water sampling

the collection of hosts throughout the year with regular periodicity. The gills of both the fishes were thoroughly examined for monogenean parasites. The gills were carefully detached, gill filaments scratched and the contents keenly observed under a stereozoom microscope (LM-52-3621Elegant). Monogenean parasites were collected with the help of small pipettes under stereozoom microscope. Since monogeneans were too small to prepare permanent slides, temporary slides were prepared using neutral red and ammonium picrate-glycerin mixture as per the method of Malmberg (1970).

Water sampling and quality analysis

Water samples were collected from the three study sites (Mylavaram, Chennur and backwaters of Somasila) at monthly interval and analyzed for various physicochemical characteristics. The temperature and pH were noted on spot, while the analysis of other parameters such as dissolved oxygen, electrical conductivity, total dissolved solids (TDS), total hardness, calcium hardness, alkalinity, chlorides (Cl^-), sulphates (SO_4^{2-}), fluorides and nitrates (NO^-_3) were carried out in Piped Water Supply (P.W.S) Laboratory located at Kadapa as per the methods of APHA (1998).

Data analysis

Ecological terminology such as infection rate, prevalence, mean intensity, mean abundance and index of infection of monogenean infections were calculated for both the fish species as per Margolis *et al*. (1982), Grabda-Kazubski *et al*. (1987) and Bush *et al.* (1997) methods, respectively. The data was subjected to Pearson's correlation using Microsoft Excel 2007 and IBM SPSS (Statistical Package for Social Sciences) 21.0 version to evaluate the relation between the environmental factors and prevalence of monogenean parasites. The present study was conducted in 3 seasons i.e., summer (March-June), rainy (July-October) and winter (November-February) and these climatic conditions were used to correlate the parasites prevalence with respect to the river water quality parameters.

RESULTS AND DISCUSSION

In present study, a total of four species of monogeneans i.e., *Cichlidogyrus sclerosus*, *Cichlidogyrus tilapiae* and *Scutogyrus longicornis* from *O. niloticus* and *Dactylogyrus fotedari* from *L. calbasu* were obtained. The prevalence and mean intensities of these four monogenean species in two hosts from the three sampling sites are depicted in Table 1. The Tables 2 - 4 show the correlation matrix between the physicochemical parameters of the three sampling stations *viz.*, Mylavaram reservoir,

Table 1: Prevalence, mean intensity and physicochemical variations in three sampling sites of river Penna, YSR Kadapa district Andhra Pradesh (India)

	Mylavaram reservoir, Mylavaram	Aadinimmayapalle dam, Chennur	Somasila backwaters, Vontimitta
Prevalence of *O. niloticus* (%)	75.0 ± 30.5	80.6 ± 24.3	73.7 ± 24.3
Mean intensity of *O. niloticus*	44.5 ± 28.6	39.7 ± 21.9	36.7 ± 22.1
Prevalence of *L. calbasu* (%)	80.0 ± 56.6	64.9 ± 45.8	85.4 ± 60.4
Mean intensity of *L. calbasu*	21.5 ± 15.2	16.1 ± 11.4	28.7 ± 20.4
<u>Water parameters</u>			
Dissolved oxygen (mg L^{-1})	7.50 ± 0.45	7.69 ± 0.34	7.84 ± 0.61
Temperature (^{o}C)	30.0 ± 4.87	29.8 ± 4.42	28.8 ± 3.65
pH	7.28 ± 0.15	7.68 ± 0.19	7.77 ± 0.10
EC (µS cm^{-1})	825.9 ± 105.7	901 ± 59.1	695.38 ± 59.7
Alkalinity (mg L^{-1})	170.84 ± 36.27	198.7 ± 24.62	261.07 ± 57.2
Total hardness (mg L^{-1})	136.46 ± 17.7	211.69 ± 39.9	256.00 ± 34.1
Calcium hardness (mg L^{-1})	70.0 ± 7.92	106.0 ± 16.53	90.2 ± 12.4
TDS (mg L^{-1})	533.6 ± 69.8	576.6 ± 37.8	431.7 ± 36.9
Chlorides (mg L^{-1})	135.6 ± 27.2	213.5 ± 29.4	109.9 ± 27.2
Fluorides (mg L^{-1})	0.23 ± 0.11	0.65 ± 0.05	0.31 ± 0.11
Sulphates (mg L^{-1})	31.5 ± 14.8	66.3 ± 12.8	20.1 ± 12.1
Nitrates (mg L^{-1})	4.12 ± 1.51	5.61 ± 0.61	2.93 ± 0.54

The values are mean ± SD

Table 2: Pearson's correlation matrix showing correlation among various physicochemical parameters of water at Mylavaram dam, Kadapa

Water parameters	DO (mg L^{-1})	Temp (^{o}C)	pH	EC (µS cm^{-1})	Alkalinity (mg L^{-1})	Total hardness (mg L^{-1})	Ca hardness (mg L^{-1})	TDS (mg L^{-1})	Chlorides (mg L^{-1})	Fluorides (mg L^{-1})	Sulphates (mg L^{-1})	Nitrates (mg L^{-1})
DO (mg L^{-1})	1.00											
Temp (^{o}C)	-0.64	1.00										
pH	0.67	-0.68	1.00									
EC (µS cm^{-1})	-0.68	0.65	-0.64	1.00								
Alkalinity (mg L^{-1})	-0.67	0.51	-0.44	0.86*	1.00							
Total hardness (mg L^{-1})	-0.36	0.69	-0.30	0.69	0.74	1.00						
Ca hardness (mg L^{-1})	-0.17	0.59	-0.14	0.65	0.63	0.94*	1.00					
TDS (mg L^{-1})	-0.70	0.66	-0.63	0.99*	0.88*	0.73	0.66	1.00				
Chlorides (mg L^{-1})	-0.68	0.57	-0.52	0.86*	0.89*	0.63	0.58	0.87*	1.00			
Fluorides (mg L^{-1})	0.60	-0.54	0.61	-0.95	-0.91	-0.68	-0.62	-0.95	-0.91	1.00		
Sulphates (mg L^{-1})	0.67	-0.39	0.52	-0.67	-0.70	-0.40	-0.28	-0.70	-0.76	0.74	1.00	
Nitrates (mg L^{-1})	0.21	-0.26	0.18	-0.42	-0.60	-0.47	-0.44	-0.44	-0.70	0.61	0.42	1.00

*Correlation is significant at $p < 0.05$ level

Table 3: Pearson's correlation matrix showing correlation among various physicochemical parameters of water at Aadinimmayepalli dam, Chennur, Kadapa

Water parameters	DO (mg L^{-1})	Temp (°C)	pH	EC (µS cm^{-1})	Alkalinity (mg L^{-1})	Total hardness (mg L^{-1})	Ca hardness (mg L^{-1})	TDS (mg L^{-1})	Chlorides (mg L^{-1})	Fluorides (mg L^{-1})	Sulphates (mg L^{-1})	Nitrates (mg L^{-1})
DO (mg L^{-1})	-											
Temp (°C)	-0.21	-										
pH	-0.24	0.53	-									
EC (µS cm^{-1})	0.23	0.44	0.46	-								
Alkalinity (mg L^{-1})	0.29	0.34	0.54	0.78	-							
Total hardness (mg L^{-1})	0.18	0.66	0.57	0.95*	0.81	-						
Ca hardness (mg L^{-1})	0.01	0.49	0.77	0.86	0.86	0.89	-					
TDS (mg L^{-1})	0.23	0.44	0.45	0.99*	0.78	0.95*	0.85	-				
Chlorides (mg L^{-1})	0.26	0.53	0.60	0.91*	0.88	0.94*	0.92*	0.91*	-			
Fluorides (mg L^{-1})	0.06	0.68	0.36	0.86	0.54	0.91*	0.72	0.86	0.78	-		
Sulphates (mg L^{-1})	-0.20	0.79	0.79	0.62	0.64	0.79	0.83	0.62	0.72	0.67	-	
Nitrates (mg L^{-1})	0.12	0.55	0.77	0.85	0.89	0.91*	0.96	0.85	0.90*	0.70	0.90*	-

*Correlation is significant at p<0.05 level

Table 4: Pearson's correlation matrix showing correlation among various physicochemical parameters of water at Somasila backwaters, Vontimitta, Kadapa

Water parameters	DO (mg L^{-1})	Temp (°C)	pH	EC (µS cm^{-1})	Alkalinity (mg L^{-1})	Total hardness (mg L^{-1})	Ca hardness (mg L^{-1})	TDS (mg L^{-1})	Chlorides (mg L^{-1})	Fluorides (mg L^{-1})	Sulphates (mg L^{-1})	Nitrates (mg L^{-1})
DO (mg L^{-1})	-											
Temp (°C)	-0.63	-										
pH	-0.32	0.25	-									
EC (µS cm^{-1})	-0.11	-0.32	-0.17	-								
Alkalinity (mg L^{-1})	-0.23	0.31	-0.17	0.04	-							
Total hardness (mg L^{-1})	0.52	-0.64	-0.64	0.46	0.18	-						
Ca hardness (mg L^{-1})	0.37	-0.66	-0.52	0.64	0.22	0.96*	-					
TDS (mg L^{-1})	-0.11	-0.32	-0.20	0.99*	0.05	0.47	0.65	-				
Chlorides (mg L^{-1})	-0.25	-0.28	0.31	0.38	0.20	-0.20	-0.06	0.34	-			
Fluorides (mg L^{-1})	-0.05	0.23	0.26	-0.44	-0.15	-0.30	-0.46	-0.43	-0.32	-		
Sulphates (mg L^{-1})	0.54	-0.49	-0.17	0.11	-0.85	0.29	0.20	0.11	-0.27	-0.09	-	
Nitrates (mg L^{-1})	-0.99	0.02	0.60	-0.26	-0.22	-0.47	-0.46	-0.26	-0.37	0.65	0.09	-

*Correlation is significant at p<0.05 level

Aadinimmayapalli dam and Somasila backwaters, respectively. Significant correlation ($p < 0.05$) was observed between electrical conductivity, total hardness, TDS, calcium hardness, alkalinity, chlorides and nitrates in the three sampling sites. Highly significant correlation existed between electrical conductivity and total hardness, TDS, chlorides and nitrates in Mylavaram reservoir; whereas Aadinimmayapalle dam exhibited highly significant correlation with alkalinity, electrical conductivity, TDS, chlorides and total hardness. Somasila backwaters demonstrated significant correlation between total hardness, TDS, electrical conductivity and calcium hardness. The dissolved

Table 5: Correlation between various physicochemical parameters of ectoparasites of *O. niloticus* in the three sampling sites, Kadapa

Water parameter ($mg\ L^{-1}$)	Mylavaram reservoir, Mylavaram						Aadinimmayepalli Dam, Chennur						Somasila backwaters, Vontimitta					
	Prevalence (%)			Mean intensity			Prevalence (%)			Mean intensity			Prevalence (%)			Mean intensity		
	CS	SL	CT	CS	SL	CT	CS	SL	CT	CS	SL	CT	CS	SL	CT	CS	SL	CT
DO	-0.17	-0.24	-0.14	-0.08	-0.35	0.17	0.27	0.24	0.32	0.32	0.22	0.36	-0.08	-0.24	-0.10	-0.42	-0.38	-0.29
Temp (°C)	0.04	0.13	0.04	0.01	0.16	-0.16	-0.73	-0.58	-0.75	-0.71	-0.62	-0.69	-0.11	0.05	-0.04	0.07	0.07	0.09
pH	-0.24	-0.29	-0.22	-0.23	-0.09	0.11	-0.37	-0.23	-0.43	-0.27	-0.25	-0.43	-0.02	0.35	-0.14	0.27	-0.05	0.37
EC ($\mu S\ cm^{-1}$)	-0.13	-0.10	-0.21	-0.25	-0.01	-0.46	-0.12	0.06	0.11	-0.05	0.02	0.01	0.48	0.32	0.52	0.30	0.27	0.29
Alkalinity	-0.24	-0.25	-0.32	-0.31	0.10	-0.42	0.05	0.06	0.06	-0.03	-0.07	0.03	0.09	0.01	0.26	-0.13	0.08	-0.07
Total hardness	-0.16	-0.13	-0.19	-0.16	-0.23	-0.27	-0.18	-0.17	-0.18	-0.20	-0.19	-0.25	0.39	0.09	0.41	-0.10	0.31	-0.15
Ca hardness	-0.20	-0.20	-0.24	-0.23	0.12	-0.27	-0.09	-0.09	-0.12	-0.06	-0.05	-0.18	0.45	0.18	0.48	-0.04	0.32	-0.05
TDS	-0.12	-0.09	-0.20	-0.24	0.05	-0.44	0.12	0.06	0.11	0.05	0.01	0.01	0.46	0.29	0.51	0.30	0.29	0.27
Chlorides	-0.31	-0.32	-0.37	-0.39	-0.02	-0.41	-0.04	0.02	-0.06	-0.02	-0.04	-0.06	0.36	0.59	0.44	0.30	0.09	0.38
Fluorides	0.25	0.26	0.35	0.34	0.08	0.52	-0.22	-0.21	-0.24	-0.22	-0.20	-0.29	-0.12	-0.11	-0.24	0.12	0.14	0.01
Sulphates	0.02	0.13	0.09	0.05	-0.23	0.07	-0.53	-0.55	-0.55	-0.51	-0.41	-0.62	0.10	0.05	-0.02	0.06	0.09	0.04
Nitrates	0.55	0.57	0.56	0.50	0.05	0.34	-0.18	-0.17	-0.18	-0.17	-0.15	-0.23	-0.33	-0.29	-0.46	0.06	-0.25	0.09

Correlation is significant at $p<0.05$ level; CS - *Cichlidogyrus sclerosus*; SL - *Scutogyrus longicornis*; CT - *Cichlidogyrus tilapiae*

oxygen recorded during the study period ranged from 7.5 to 7.85 mg L^{-1} at temperatures ranging between 28.8 - 30.0°C (Table 1) which indicates the safety limits for freshwater ecosystem (Chapman *et al*., 2000). The data from these sites showed that the fishes were infected in almost all the sampling locations with highest infection noticed in Somasila backwaters for *D. fotedari* in *L. calbasu* (85.4%) and Aadinimmayapalle dam for Cichlidogyrids in *O. niloticus* (80.6%) which might be due to the variations in the physicochemical parameters because of the accumulation of different ions such as chlorides, fluorides, nitrates, calcium etc., by various human anthropogenic activities in these two sites. The alkalinity was found within the acceptable range (50-300 mg L^{-1} $CaCO_3$) in the three sites with highest values in Somasila backwaters (261.1 mg L^{-1} $CaCO_3$).

Table 5 shows the correlation matrix between monogenean prevalence and water quality variables of the three study sites. The prevalence and mean intensity of all the three monogeneans established a significant positive correlation with nitrates (0.55, 0.56 and 0.57) and fluorides (0.25, 0.35 and 0.26); while the remaining parameters showed negative insignificant correlation with the occurrence of three parasites in Mylavaram reservoir. Similarly, the prevalence and mean intensity of all the three monogeneans exhibited a significant negative correlation with temperature and nitrates. Except the dissolved oxygen, all the other parameters showed a negative correlation with the occurrence of these parasites in Aadinimmayapalli dam. However, the prevalence and mean intensity of three monogeneans exhibited moderate positive correlation with electrical conductivity, calcium hardness, TDS and chlorides and the remaining parameters showed negligible positive and negative correlation with the occurrence of the three parasites (Table 5). The prevalence and mean intensity of *D. fotedari* obtained from *L. calbasu* established a positive correlation with total hardness (0.53), alkalinity (0.41), calcium hardness (0.36), total dissolved salts (0.30) and temperature (0.302) while the other parameters showed insignificant positive and negative correlations with the prevalence and mean intensity of *D. fotedari* in Mylavaram reservoir. The prevalence and mean intensity of *D. fotedari* showed insignificant negative correlation with all the12 physicochemical parameters of water samples collected from Aadinimmayapalle dam. In Somasila backwaters, the prevalence of *D. fotedari* established the correlation with calcium hardness (0.54), total hardness (0.49) and TDS (0.74) while the rest of the parameters showed negligible positive correlation or negative correlation with the occurrence of *D. fotedari*. Similarly, the mean intensity of *D. fotedari* showed very negligible correlation with the physicochemical parameters (Table 6). The infestation rates of *D. fotedari* was high in fishes collected from back waters of Somasila reservoir with prevalence and mean intensity of 85.4% and 28, respectively and least in fishes collected from Mylavaram reservoir (Table 7). The infestation rate of *Cichlidogyrids* was high in the fishes collected from Aadinimmayapalle dam with the prevalence and mean intensity of 80.6% and 39.7, respectively, and least in fishes collected from backwaters of Somasila reservoir (P = 73.7%; MI = 36.7) (Table 7).

Table 6: Correlation between the various physicochemical parameters of water at various sampling sites in Kadapa and *D. fotedari* (the ectoparasite of *L. calbasu*)

Water parameters	Mylavaram		Aadinimmayapalle Dam, Chennur		Somasila backwaters, Vontimitta	
	Prevalence (%)	Mean intensity	Prevalence (%)	Mean intensity	Prevalence (%)	Mean intensity
DO (mg L^{-1})	-0.47	-0.47	-0.13	-0.24	0.22	-0.17
Temperature (°C)	0.30	0.30	-0.29	-0.22	-0.46	-0.18
pH	-0.12	-0.12	-0.03	-0.24	0.08	0.31
EC (µS cm^{-1})	0.23	0.23	-0.35	-0.36	0.37	0.34
Alkalinity (mg L^{-1})	0.41	0.41	-0.49	-0.69	-0.029	-0.002
Total hardness (mg L^{-1})	0.53*	0.53	-0.41	-0.42	0.49	0.08
Ca hardness (mg L^{-1})	0.36	0.36	-0.27	-0.40	0.54*	0.19
TDS (mg L^{-1})	0.30	0.30	-0.35	-0.36	0.37	0.34
Chlorides (mg L^{-1})	0.31	0.31	-0.43	-0.54	-0.23	-0.02
Fluorides (mg L^{-1})	-0.21	-0.21	-0.38	-0.19	0.02	0.09
Sulphates (mg L^{-1})	-0.29	-0.29	-0.16	-0.25	0.29	-0.001
Nitrates (mg L^{-1})	-0.30	-0.30	-0.28	-0.48	0.18	0.31

The variation in the prevalence of monogenean infection with changes in water quality and the incidence of infection can be reflected with seasons in both the fishes i.e., higher during winter season and lowest during summer (Table 8). *O. niloticus* showed a significant difference in the prevalence and mean intensities of *Cichlidogyrids* between the seasons (analysis of variance, F = 4.21, p = 0.017). Only the parasitism in rainy and winter showed a statistically significant difference (t = -2.89, p = 0.002). Parasitosis was highest in winter season (47%), followed by rainy season (32%) and summer season (22%) [Table 8]. However, *L. calbasu* showed a significant difference in the prevalence and mean intensities of *D. fotedari* between the seasons (analysis of variance, F = 3.69, p = 0.029). Only the parasitisation in summer and winter seasons exhibited a statistically significant difference (t = -5.30, p = <0.000). Parasitisation was highest in winter (93.2%), followed by rainy (87.5%) and least in summer seasons (42.1%) [Table 8].

The alarming environmental degradation can be better understood and analyzed by the use of biological tags as they can sense the changes in the environment with changes in fish parasite number,

Table 7: Prevalence and mean intensity of monogenean species recorded from two fish species at the three sampling locations of river Penna in YSR district, Kadapa

Collection sites	Total number of fishes	Infected fishes	Total number of parasites	Prevalence (%)	Mean intensity	Mean abundance
Cichlidogyrus sclerosus						
Chennur	36	29	673	80.56	23.21	18.69
Mylavaram	40	28	696	70.00	24.86	17.40
Somasila	57	42	876	73.68	20.86	15.37
Scutogyrus longicornis						
Chennur	36	26	334	72.22	12.85	9.28
Mylavaram	40	25	377	62.50	15.08	9.43
Somasila	57	32	347	56.14	10.84	6.09
Cichlidogyrus tilapiae						
Chennur	36	17	143	47.22	8.41	3.97
Mylavaram	40	24	262	60.00	10.92	6.55
Somasila	57	35	320	61.40	9.14	5.61
Dactylogyrus fotedari						
Chennur	57	37	598	64.90	16.10	10.49
Mylavaram	10	08	172	80.00	21.50	17.20
Somasila	55	47	1353	85.40	28.70	24.60

Table 8: Seasonal variation in the prevalence (%) and mean intensity of *D. fotedari* in *L. calbasu* and *Cichlidogyrids* in *O. niloticus*

Parasite species	Seasons	Fishes examined (No.)	Prevalence (%)	MI	F	p-value ($p < 0.05$)	Comparison two by two	t-test	p-value ($p < 0.05$)
D. fotedari	Summer	38	42.1	14			Summer-Rainy	-1.54	0.063
	Rainy	40	87.5	23	3.69	0.029	Summer -Winter	-5.30	<0.00*
	Winter	44	93.2	26.7			Rainy-Winter	-0.533	0.29
Cichlidogyrids	Summer	43	22	935			Summer-Rainy	-1.42	0.079
	Rainy	40	32	1201	4.21	0.017*	Summer -Winter	-1.37	0.086
	Winter	50	47	1892			Rainy-Winter	-2.89	0.002*

*= Significant level at $p< 0.05$; MI- Mean intensity; F - Anova test;

physiology or chemical composition. Fish parasites, especially ectoparasites (monogeneans, isopods and copepods) are used as potential bioindicators all over the world due to their direct contact with the host and environment. The results clearly specify that there were significant differences between monogenean prevalence and water quality parameters in fish gills with respect to the location and seasons. The highest prevalence was noticed during winter followed by rainy season which could be attributed to the higher water quality variables of river like temperature. Temperature is considered as one of the most significant abiotic factors of aquatic ecosystem and all the aquatic organisms depend on water temperatures for their physiological processes (Simkova *et al*., 2001; Modu *et al*., 2014). Higher temperatures speed up chemical activity in the organisms, reduction in gas solubility, intensify odour and taste of water (Simkova *et al*., 2001). Variation in monogenean prevalence in the fishes can be attributed to changes in the temperature in different seasons (Modu *et al*., 2012). The lower temperatures during the winter and rainy seasons are suitable for the recruitment of monogenean parasites. However, Rakauskas and Blaeviéius (2010) are of the opinion that the other abiotic factors like DO, pH, alkalinity, TDS, calcium hardness, chlorides and nitrates apart from temperature, and biotic factors can influence the proliferation of monogeneans and facade the effect of temperature. Similarly, DO in aquatic ecosystems is deemed to be an essential abiotic factor. Low and high DO due to pollution in water may make the life of aquatic organisms vulnerable (Molnar, 1994). In present study, all the three sampling sites exhibited DO levels within the safety limits. Mylavaram reservoir showed lowest levels of monogenean infection, oxygen rich water seems to support the health conditions and immune system of fish hosts leading to the reduction in the proliferation of monogenean infection. However, the other sampling sites seems to be more degraded and polluted by human activities thereby revealing the changing water quality and thus paving the pathway for more monogenean infestations.

Hydrogen ion concentration (pH) seems to have no impact on infection variables of monogeneans during the current study as the three sites didn't show marked variations in pH which correlates with those of Biswas and Pramanik (2016) and El-Naggar *et al*. (2017). The other constituents such as total dissolved solids, total hardness, electrical conductivity, calcium hardness, chlorides, sulphates, nitrates and alkalinity were more in polluted environments- Aadinimayyapalli dam and Somasila backwaters as these two sampling sites are downstream to Mylavaram reservoir. As the river approaches towards sea, it gathers more pollutants before merging into the sea. Hence, the poor quality of water in aquatic ecosystem such as Aadinimayyapalli dam and Somasila backwaters is causing traumatic situations that deteriorates immune system and change resistance strategy of the aquatic organisms, leading to poor fish health. The study revealed that the monogeneans can acts as efficient biological indicators.

Conclusion: This study revealed that the monogeneans are highly host specific parasites in fishes. Their direct life-cycles, host specificity and direct contact with the external environment made them exceptional biological indicators. These types of studies are extremely necessary from time to time to know the extent of pollution in a particular water body and take necessary steps to reduce the pollution levels of the water body. Also, the study revealed that monogeneans can serve as excellent biological tags. Therefore, understanding their ecological aspects under natural conditions within the host can provide a baseline data for the benefit of these two fish stocks in the extensive aquaculture.

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